Depth-Three Smith Profiles and Newton-Minor Geometry for Affine Hjelmslev–Radon Incidence

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Abstract

Let B_3 be the integer line-by-point incidence matrix of the affine plane over $\mathbb{Z}/p^3\mathbb{Z}$, with primitive normal directions taken modulo units. We determine every nonzero Smith factor for every prime. The exponent is p^5, and the six multiplicities $m_0, \ldots, m_5$ are given by explicit polynomials on the two residue classes $p \equiv 1, 5 \pmod 6$, together with separate exact rows for $p = 2, 3$. The new ingredient is a reduction of the first unresolved integral obstruction to a truncated q-Pascal common image. An exact q-Lucas block product identifies its supported columns; a scalar recurrence proves module closure and boundary generation; and a q-Newton row basis makes all determinantal ideals triangular with Smith thresholds

$$\tau_p(K) = \mu_p\bigl(p^2(p-1) - K\bigr) + p\beta_p(K).$$

This also yields the complete ambient filtered profile, Hilbert series and minimal number of generators of the common image. A typed cross-chart exchange closes the remaining transfer defect. Finite computations are used only for exceptional certificates and regression, not to interpolate the all-prime formulas.

Keywords: Smith normal form; affine Hjelmslev geometry; Radon transform; incidence matrix; residue ring; q-Pascal matrix; Newton minors.

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Companion repository and reproducibility materials

1 Contribution and logical spine

The incidence object in this paper is affine and integral. It is neither a projective incidence matrix nor a matrix over a field having the same cardinality as $\mathbb{Z}/p^3\mathbb{Z}$. The main result, theorem 9.1, gives the complete Smith profile at depth three for every prime. Its proof is not a finite-prime fit. It combines four uniform layers:

1. a maximal-order and point-fibre transfer analysis, which fixes the total determinant order and high Smith tail;

2. a complete-ideal calculation of the modular rank;
3. an odd-prime tripotent-sector reduction to a truncated q-Pascal common image, followed by an exact Newton-minors calculation of that image;
4. a cross-chart divided-carry theorem, which removes the final transfer defect before the exact Smith bookkeeping is applied.

The written proof and the finite calculations play distinct roles. The exceptional primes 2 and 3 have exact finite certificates where a semisimple or degree-bound argument does not apply. The primes 5, 7, and 11 are retained as public exact regression rows evaluated from the tracked formulas. They do not enter the selection of either residue-class branch, and none of these finite rows proves a uniform polynomial.

The diagonal cyclic valuation formulas used in the top-shell calculation belong to Weber–Künzer [9]; the cyclic CRT layer is also covered by Mahatab–Sampath [15]. The complete-ideal correspondence and Hoskin–Deligne length formula are imported through Lipman's work [13, 14]. Sagan supplies q-Lucas [17], while the integral q-Stirling expansion is the Carlitz identity in the form recorded by Cai–Ehrenborg–Readdy [3]. These sources do not state the affine common-image theorem, the Newton-minors theorem, or the profile in theorem 9.1.

Sections 2–3 fix the structural invariants. Sections 4–7 compute the first integral obstruction. Section 8 closes the cross-chart transfer defect, and Sections 9–10 assemble and check the answer. Section 11 defines the public reproducibility and repository boundary.

2 Affine incidence lattice, maximal order, and transfer

Definition 2.1 (Affine Hjelmslev–Radon incidence)**.** Fix a prime p and $n \geq 1$, and put

$$\mathsf{R}_n = \mathbb{Z}/p^n\mathbb{Z}, \qquad G_n = \mathsf{R}_n^2.$$

A vector $u = (a, b)$ is primitive if $a\mathsf{R}_n + b\mathsf{R}_n = \mathsf{R}_n$. Primitive vectors are identified under multiplication by $\mathsf{R}_n^\times$, and we use the chart

$$\mathbb{P}^1(\mathsf{R}_n) = \{(1, t) : t \in \mathsf{R}_n\} \sqcup \{(ps, 1) : s \in \mathbb{Z}/p^{n-1}\mathbb{Z}\}.$$

For $u \in \mathbb{P}^1(\mathsf{R}_n)$ and $c \in \mathsf{R}_n$, let $\ell_{u,c} = \{x \in G_n : u \cdot x = c\}$. The matrix B_n has rows (u, c), columns $x \in G_n$, and entries

$$B_n[(u, c), x] = \mathbf{1}_{\{u \cdot x = c\}}.$$

Thus B_n has $(p + 1)p^{2n-1}$ rows, p^{2n} columns, and each row has p^n ones.

Let $\mathsf{A}_n = \mathbb{Z}[G_n]$. For $H_u = \ker(u)$ put $N_{H_u} = \sum_{h \in H_u}[h]$ and

$$\mathsf{L}_n = \sum_u \mathsf{A}_n N_{H_u}, \qquad \mathsf{Q}_n = \mathsf{A}_n/\mathsf{L}_n.$$

Lemma 2.2 (Group-ring dictionary)**.** *There is a canonical isomorphism*

$$\mathsf{Q}_n \cong \operatorname{coker}(B_n^{\mathsf{T}}).$$

In particular, Q_n *records the* p^{2n} *nonzero Smith factors of* B_n*.*

Proof. For fixed u, the p^n affine fibres are disjoint translates of H_u; their indicators form a $\mathbb{Z}$-basis of $\mathsf{A}_n N_{H_u}$. Summing over the direction chart identifies the image of B_n^{T} with L_n. $\square$

The rational characters of exact order p^k form Galois orbits indexed by cyclic subgroups of order p^k. These subgroups form a rooted projective character tree; level k has $(p+1)p^{k-1}$ nodes, each carrying $\mathbb{Z}[\zeta_{p^k}]$.

Theorem 2.3 (Maximal-order quotient)**.** *Let* $\mathcal{O}_n$ *be the maximal order in* $\mathbb{Q}[G_n]$*. Then*

$$\mathcal{O}_n/\mathsf{L}_n \cong \mathbb{Z}/p^{2n}\mathbb{Z} \oplus \bigoplus_{k=1}^{n} (\mathbb{Z}/p^{2n-k}\mathbb{Z})^{p^{2k}-p^{2k-2}}, \tag{2.1}$$

$$v_p|\mathsf{Q}_n| = \frac{np^{2n}}{2} - \frac{p(p^{2n}-1)}{2(p^2-1)}. \tag{2.2}$$

In particular,

$$v_p|\mathsf{Q}_3| = \frac{3p^6 - p^5 - p^3 - p}{2}.$$

Proof. For a leaf C in the character tree let R_C be the image of the cyclic group ring on the root-to-leaf components and put $\mathsf{M}_n = \sum_C \mathsf{R}_C$. Fourier evaluation of a line norm gives the exact additive identity $\mathsf{L}_n = p^n \mathsf{M}_n$. Projection to the top cyclotomic level is onto. On each branch its kernel is generated by $\Phi_{p^n}(t)$, whose value on every lower p-power root is p. Top components are leaf-disjoint, hence

$$\mathsf{M}_n \cap \mathcal{O}_{<n} = p\mathsf{M}_{n-1}.$$

Induction gives exponent n once at the root and exponent $n-k$ with multiplicity $p^{2k} - p^{2k-2}$ at level k for $\mathsf{M}_n \subset \mathcal{O}_n$. Scaling by p^n proves (2.1). Subtracting the standard group-ring normalization index from the exponent sum yields (2.2). The cyclic branch kernel is the prior-art CRT layer [15]; the projective-tree assembly is the step used here. $\square$

Let $\pi_n : G_n \to G_{n-1}$ be reduction and define point-fibre transfer

$$\tau_n([a]) = \sum_{\pi_n(x)=a} [x].$$

Theorem 2.4 (Exact transfer and multiscale pencil)**.** *For* $n \geq 2$*,*

$$\tau_n(\mathsf{A}_{n-1}) \cap \mathsf{L}_n = \tau_n(\mathsf{L}_{n-1}),$$

so τ_n *induces an injection* $\mathsf{Q}_{n-1} \hookrightarrow \mathsf{Q}_n$*. Moreover, in* Q_n*,*

$$p^n[x] = \tau_n\big(p^{n-2}[x \bmod p^{n-1}]\big), \qquad p^n\mathsf{Q}_n = \tau_n(p^{n-2}\mathsf{Q}_{n-1}). \tag{2.3}$$

Consequently $\exp(\mathsf{Q}_n) = p^{2n-1}$ *and, if* $m_h^{(n)}$ *denotes the multiplicity of* $\mathbb{Z}/p^h\mathbb{Z}$ *in* Q_n*, then*

$$m_{n+h}^{(n)} = m_{n-2+h}^{(n-1)} \qquad (1 \leq h \leq n-1). \tag{2.4}$$

Proof. In Fourier coordinates a point fibre has value p^2 on lower characters and zero on exact-order-p^n characters. Combining this with the leafwise description in theorem 2.3 gives

$$\tau_n(a) \in \mathsf{L}_n \iff a \in \mathsf{L}_{n-1},$$

and disjoint point-fibre supports give injectivity.

For the second assertion let $F_{r,n}(x)$ be the sum of points congruent to x modulo p^r, and let S_x be the sum of the unique line of every direction through x. Coefficient counting by the valuation of $y-x$ gives in A_n

$$S_x = F_{0,n} + \sum_{r=1}^{n-1}(p^r - p^{r-1})F_{r,n}(x) + p^n[x].$$

Both S_x and $F_{0,n}$ vanish in Q_n. Subtracting the transferred depth-$(n-1)$ identity leaves (2.3). Injection and induction from the depth-one affine profile [4] give the exact exponent; comparing cyclic factors in (2.3) gives (2.4). □

At depth two write

$$\alpha = \frac{p^3(p-1)}{2}, \qquad \beta = \frac{p(p-1)^2(p+2)}{4}, \qquad \gamma = \frac{p(p-1)}{2}. \tag{2.5}$$

The depth-two profile has multiplicities α, β, γ at exponents $1, 2, 3$ [1]. Hence (2.1)–(2.4) imply for depth three

$$m_4 = \beta, \qquad m_5 = \gamma, \tag{2.6}$$

and exclude exponents above five.

3 The modular rank and the cyclic top-shell count

We next determine $m_0 = \operatorname{rank}_{\mathbb{F}_p} B_3$. Let $\overline{\mathsf{A}}_3 = \mathbb{F}_p[(\mathbb{Z}/p^3\mathbb{Z})^2]$ and let I_3 be the span of all affine-line indicators. In augmentation coordinates $\overline{\mathsf{A}}_3 \cong k[x,y]/(x^{p^3}, y^{p^3})$, put $J_3 = \operatorname{Ann}(I_3)$.

Lemma 3.1 (Depth-three bundle annihilator)**.** *For each $e \in \mathbb{P}^1(\mathbb{Z}/p^2\mathbb{Z})$ choose regular parameters (z_e, v_e) along the associated direction. Then*

$$J_3 = \bigcap_e (z_e, v_e^{p^2})^p. \tag{3.1}$$

Proof. A primitive cyclic line subgroup $H = \langle h \rangle$ has norm $N_H = (h-1)^{p^3-1}$ in characteristic p, hence $\operatorname{Ann}(N_H) = (h-1)$. Group the p fine lifts above a depth-two direction as $h_{e,a} = g_e t_e^{p^2 a}$. With $z = g_e - 1$ and $w = v^{p^2}$, their linear factors are $(1+z)(1+w)^a - 1$. Modulo $(z,w)^p$, truncated logarithms send them to the p distinct forms $Z + aW$. A polynomial of degree below p divisible by all these forms is zero, so the intersection contains no nonzero class of degree below p. Conversely, modulo each monic factor $(1+z)(1+w)^a - 1$ one has $z \in (w)$; hence every monomial of total degree p maps into $(w^p) = 0$. Thus $(z,w)^p$ lies in every factor ideal, and their intersection is exactly $(z,w)^p$. Intersecting all bundles gives (3.1). □

Lemma 3.2 (Formal lift and simultaneous base-point cluster). *Put $N = p^3$, $q = p^2$, let $R = k[[x, y]]$ with maximal ideal $\mathfrak{m} = (x, y)$, and choose formal lifts of the regular pairs in lemma 3.1. Set*

$$\widetilde{I}_e = (z_e, v_e^q)^p, \qquad \widetilde{J}_3 = \bigcap_{e\in\mathbb{P}^1(\mathbb{Z}/p^2\mathbb{Z})} \widetilde{I}_e.$$

Then $\mathfrak{m}^N \subseteq \widetilde{J}_3$, the ideal $\widetilde{J}_3$ is complete, and reduction modulo (x^N, y^N) identifies it with J_3. A common point-blowup resolution has one root, $p+1$ trunks of length $p-1$, and p branches above each trunk of length

$$L = q - p = p(p-1).$$

In the strict exceptional basis its componentwise-maximum divisor has coefficients

$$p \quad \text{at the root,} \qquad p(j+1) \quad (1 \le j \le p-1) \quad \text{on a trunk,}$$

and

$$p(p+r) \quad (1 \le r \le L) \quad \text{on a branch.}$$

Its least antinef majorant represents $\widetilde{J}_3$. In particular,

$$\operatorname{rank}_k B_3 = \dim_k I_3 = \dim_k(\overline{\mathsf{A}}_3/J_3) = \operatorname{length}_R(R/\widetilde{J}_3).$$

Proof. In any regular pair (z, v), a monomial $z^a v^b$ outside $(z, v^q)^p$ satisfies $a+\lfloor b/q\rfloor \le p-1$, hence $a + b \le pq - 1 = N - 1$. Therefore $\mathfrak{m}^N$ lies in every $\widetilde{I}_e$. The same monomial description says

$$(z, v^q)^p = (z^a v^b : a, b \ge 0,\ qa + b \ge pq),$$

whose exponent set is exactly the set of lattice points in its Newton polyhedron. Hence each $\widetilde{I}_e$ is integrally closed, and their finite intersection is integrally closed as well: an element integral over the intersection is integral over every $\widetilde{I}_e$, and hence belongs to every $\widetilde{I}_e$. Thus $\widetilde{J}_3$ is a complete $\mathfrak{m}$-primary ideal. Since $(x^N, y^N) \subseteq \mathfrak{m}^N \subseteq \widetilde{I}_e$ for every e, finite intersection commutes with the Artinian quotient and

$$\widetilde{J}_3/(x^N, y^N) = J_3, \qquad \operatorname{length}_R(R/\widetilde{J}_3) = \dim_k(\overline{\mathsf{A}}_3/J_3).$$

Let $\bar{k}$ be an algebraic closure of k. Because all ideals contain $\mathfrak{m}^N$, the base change $R \to \bar{k}[[x, y]]$ may be computed in the finite quotient $R/\mathfrak{m}^N$. Faithful flatness makes it commute with the finite intersection and gives

$$\dim_k(R/\widetilde{J}_3) = \dim_{\bar{k}}\big(\bar{k}[[x, y]]/\widetilde{J}_3\bar{k}[[x, y]]\big).$$

We may therefore use the complete-ideal/antinef correspondence over $\bar{k}$ and descend its numerical colength to k.

We now construct the common cluster. In the finite-slope chart write a direction as $a = a_0 + pa_1$, with $a_0, a_1 \in k$, and take

$$z_a = (1 + x)^{-a}(1 + y) - 1, \qquad v_a = x.$$

For two lifts of the same coarse direction, Frobenius gives

$$1 + z_{a_0+pa_1} = (1 + z_{a_0+pb_1})(1 + x^p)^{\,b_1-a_1}.$$

If $a_1 \neq b_1$, reduction modulo $z_{a_0+pb_1}$ has first nonzero term $(b_1 - a_1)x^p$. The two smooth branches therefore have intersection multiplicity exactly p. After j blowups along their common smooth branch, $0 \leq j < p$, their difference has leading term a nonzero multiple of x^{p-j}; after the pth it records the distinct residue a_1. They therefore share the root and $p-1$ successive free infinitely-near points, and then separate at the p distinct values of a_1. Directions with different a_0 have distinct linear tangents $y - a_0 x$ and separate after the first blowup. Interchanging x and y treats the remaining chart $(pa_1, 1)$. Hence there are exactly $p+1$ coarse trunks, with p branches over each trunk. Every center lies on the strict transform of a smooth branch and only on the newest exceptional component, so no center is satellite. The coarse tangents, the separation parameters a_1 and all subsequent centers are k-rational; hence every residue-degree factor in the Hoskin–Deligne sum is one. For $p = 2$ the same construction has three trunks of length one and two branches of length two above each trunk; the coefficient $b_1 - a_1$ remains nonzero, so no contact or separation step degenerates.

For one direction, follow the smooth branch $z_e = 0$. At the jth center, successive substitutions $z_{e,j-1} = v_e z_{e,j}$ give the weak transforms

$$(z_{e,j}, v_e^{q-j})^p, \qquad 0 \leq j < q.$$

Thus $\widetilde{I}_e$ has point-basis multiplicity p at the root and at each of a free chain of $q-1$ further points. Its first p centers are the root and the common trunk; after separation the remaining $q - p = L$ centers form its branch.

Let $\mu : X \to \operatorname{Spec} R$ be the common resolution and let D_e be the divisor of $\widetilde{I}_e$ on X. Along its own free chain, point-basis multiplicity p makes the coefficient of the exceptional component created at depth d in the strict exceptional basis equal to $p(d+1)$. An ideal whose branch has already separated has unit weak transform at a center on another branch, so further pullback only propagates its coefficient at the last common component; it cannot exceed the coefficient supplied by the ideal belonging to that branch. Taking $D^{\mathrm{raw}} = \max_e D_e$ componentwise therefore gives exactly the displayed root, trunk and branch coefficients. A rational function is a local section of $\mathcal{O}(-D^{\mathrm{raw}})$ precisely when its exceptional valuations dominate every D_e; hence $\mathcal{O}(-D^{\mathrm{raw}}) = \bigcap_e \mathcal{O}(-D_e)$ and

$$\mu_* \mathcal{O}(-D^{\mathrm{raw}}) = \bigcap_e \mu_* \mathcal{O}(-D_e) = \bigcap_e \widetilde{I}_e = \widetilde{J}_3.$$

Unloading replaces D^{raw} by its unique least antinef majorant without changing this pushforward [13, Section 1][14, Corollary 1.1.2 and Lemma 1.2]. Finally, affine translation preserves the line span, so I_3 is an ideal. The coefficient-of-identity pairing makes $\overline{\mathsf{A}}_3$ a symmetric Frobenius algebra; its nondegeneracy therefore gives $J_3 = \operatorname{Ann}(I_3) = I_3^{\perp}$, and the final displayed equality follows. $\square$

The preceding lemma supplies the raw divisor rather than assuming it. Symmetry and unloading now give the following least antinef point basis.

Theorem 3.3 (Depth-three modular rank)**.** *Let A be the root point-basis coefficient, H the common point-basis coefficient at every trunk point and $K_1, \ldots, K_L$ the successive* *point-basis coefficients on each branch. The least antinef majorant has point basis*

case	A	H	$(K_i)_{1\le i\le L}$
$p=3$	15	3	$1,\dots,1$
$p=3h+1$	$(p+1)H+p$	$\dfrac{(p-1)(p+2)}{3}$	$h,\dots,h$
$p=3h+2$	$(p+1)H$	$\dfrac{p(p+1)}{3}$	$h+1$ (*ph times*), h ($L-ph$ *times*).

(3.2)

The last line includes $p=2$*. Consequently*

$$M_p := \operatorname{rank}_{\mathbb{F}_p} B_3 = \begin{cases} 240, & p=3,\\ \frac{p(p+1)(3p^4+4p^3+3p-1)}{18}, & p\equiv 1 \pmod 3,\\ \frac{p^2(p+1)^2(3p^2+p+1)}{18}, & p\equiv 2 \pmod 3. \end{cases} \tag{3.3}$$

Proof. The least antinef majorant is unique. The decorated tree is invariant under permutations of equal trunks and of the p branches above each trunk, so the least majorant has a symmetric point basis. Write h for its common trunk coefficient at the first trunk point, and write $H_1,\dots,H_{p-1}$ for the successive trunk coefficients, so $H_1=h$. On each branch its point-basis coefficients satisfy

$$h=H_1\ge H_2\ge\cdots\ge H_{p-1}\ge pK_1, \qquad K_1\ge K_2\ge\cdots\ge K_L\ge 0,$$

while root proximity gives $A\ge (p+1)h$. The corresponding strict exceptional coefficients are

$$A+\sum_{q=1}^{j} H_q \quad (1\le j\le p-1), \qquad A+\sum_{q=1}^{p-1} H_q+\sum_{i=1}^{r} K_i \quad (1\le r\le L).$$

The last one must dominate the raw endpoint coefficient p^3. Since $H_q\le h$ and $K_i\le \lfloor h/p\rfloor$, every symmetric antinef majorant therefore satisfies

$$A\ge \max\left((p+1)h, p^3-(p-1)h-p(p-1)\left\lfloor \frac{h}{p}\right\rfloor\right). \tag{3.4}$$

Put $F_p(h)$ equal to the right-hand side. Writing $h=pk+s$, $0\le s<p$, makes the second entry minus the first

$$p\big(p^2-(3p-1)k-2s\big).$$

If $p=3m+1$, the unique minimum occurs at $k=m$, $s=2m$, namely $h=(p-1)(p+2)/3$: the difference is p there and $-p$ at $h+1$, while the decreasing entry is larger by $p-1$ at $h-1$. Thus $F_p(h)=(p+1)h+p$. If $p=3m+2$, the unique minimum is at $h=p(m+1)$; comparison with $h-1$ and $h+1$ gives $F_p(h)=(p+1)h$. For $p=3$, direct evaluation gives the unique minimum $F_3(3)=15$.

It remains to show that the lower bounds are attained in the proximity partial-sum order. In the $p=3m+1$ and $p=3$ cases, equality at the terminal raw coefficient forces all $H_q=h$ and every branch coefficient to be m and 1, respectively. In the $p=3m+2$ case, attaining the lower bound in (3.4) likewise forces $H_q=h$ for all q; otherwise the terminal strict coefficient would fall below p^3. The branch sum must then be at least

$$p^3-A-(p-1)h=Lm+pm.$$

Among nonincreasing integer sequences of length L, bounded above by $m+1$, and with sum at least $Lm+pm$, the sequence with pm entries $m+1$ followed by $L-pm$ entries m has the smallest prefix sums. It is therefore the least sequence in the order imposed by proximity. Substitution in $A+\sum_{q\le j} H_q$ and in the branch prefix sums verifies every intermediate raw bound $p(j+1)$ and $p(p+r)$, with equality at the endpoint. Hence no coefficient of any antinef majorant can be lowered, proving (3.2).

The Hoskin–Deligne formula [13, 14] now gives

$$M_p = \binom{A+1}{2} + (p+1)(p-1)\binom{H+1}{2} + (p+1)p\sum_{i=1}^{L}\binom{K_i+1}{2}. \tag{3.5}$$

Substitution and simplification yield (3.3). For $p=3$, this is $120+48+72=240$; no finite profile is used in the derivation. ☐

Corollary 3.4 (First transfer congruence)**.** *For every prime p,*

$$\tau_3(\mathsf{A}_2) \subseteq \mathsf{L}_3 + p\mathsf{A}_3.$$

Proof. Under the coefficient pairing, the displayed inclusion is equivalent to $J_3 \subseteq (x^{p^2}, y^{p^2})$. For $p\ge 5$, the augmentation order of the complete ideal J_3 is the root point-basis coefficient A in (3.2), and those formulas give $A \ge 2p^2-1$ for $p\ge 5$. A monomial whose two exponents are both below p^2 has total degree at most $2p^2-2$; hence every element of J_3 lies in (x^{p^2}, y^{p^2}). For $p=2,3$, the two-lane exact certificate in section C constructs one point-fibre indicator in the modular depth-three line code and independently verifies its orthogonality to the full dual nullspace. Affine translation gives all point fibres, which is the same inclusion. Thus the congruence holds for every prime. ☐

The first cyclic diagonal block in the relative top shell has a unit count a_0 which must not be confused with $m_0 = M_p$.

Proposition 3.5 (Cyclic unit count)**.**

$$a_0 = \begin{cases} 8, & p=2, \\ 108, & p=3, \\ \dfrac{(p-1)(3p^5+2p^4+2p^2+2)}{18}, & p>3,\ p\equiv 1 \pmod 3, \\ \dfrac{p(p+1)(12p^4-16p^3+10p^2-3p+1)}{72}, & p\equiv 2 \pmod 3. \end{cases} \tag{3.6}$$

Proof. Write a cyclic index as $j=a+bp+cp^2$. The Weber–Künzer valuation [9] becomes

$$\beta_p(j) = a + (2p-1)b + p(3p-2)c.$$

The unit count is the sum of $p^2(p-1)-\beta_p(j)$ over the region where this quantity is positive. For $p=6h+1$, the region consists of $0\le c\le 2h-1$ with all a,b, followed by the two boundary slices $c=2h$ and $b=2h$. For $p=6h+5$, it consists of $0\le c\le 2h$ with all a,b and the slice $c=2h+1$, $0\le b\le 5h+3$. Summing these arithmetic progressions gives (3.6); $p=2,3$ are the degenerate boundary cases and are evaluated directly. ☐

4 Tripotent sectors and the reduced q-Pascal common image

Fix a prime p. Put

$$k = \mathbb{F}_p, \qquad N = p^3, \qquad N_0 = p^2, \qquad e = N - N_0, \qquad \overline{\mathcal{O}} = k[s]/(s^e), \qquad \zeta = 1 + s.$$

The phrase "q-Pascal" is a conventional name. Whenever a Gaussian polynomial variable is required below it is denoted by $\mathfrak{q}$; the letters r, t, u denote residue-class labels and h denotes an ordinary integer in a congruence decomposition of p. Thus no residue integer is also the q-Pascal indeterminate.

Define the source and the two different chart targets by

$$\begin{aligned} U_N^X &= k[X]/((X-1)^e), & R_N^Y &= k[Y]/((Y-1)^N), \\ U_N^Y &= k[Y]/((Y-1)^e), & D_1 &= U_N^X \otimes_k R_N^Y, \\ E_{\mathcal{A}} &= \overline{\mathcal{O}}^{\mathbb{Z}/N\mathbb{Z}}, & E_{\mathcal{X}} &= \overline{\mathcal{O}}^{\mathbb{Z}/N_0\mathbb{Z}}. \end{aligned}$$

The modular chart maps have the explicit types

$$\begin{aligned} \mathcal{A} &: D_1 \longrightarrow E_{\mathcal{A}}, & \mathcal{A}(b)_t &= b(\zeta, \zeta^t), & t &\in \mathbb{Z}/N\mathbb{Z}, \\ \mathcal{X} &: D_1 \longrightarrow E_{\mathcal{X}}, & \mathcal{X}(b)_u &= b(\zeta^{pu}, \zeta), & u &\in \mathbb{Z}/N_0\mathbb{Z}. \end{aligned} \tag{4.1}$$

Let $\rho_Y : R_N^Y \twoheadrightarrow U_N^Y$ be truncation and let $\iota_Y : U_N^Y \to R_N^Y$ be the degree-below-e k-linear section. Put

$$\pi = \mathrm{id} \otimes (\iota_Y \rho_Y), \qquad V_0 = \ker \pi, \qquad E = \operatorname{im} \pi = U_N^X \otimes_k U_N^Y.$$

On E let τ exchange the two truncated tensor factors, and extend it to the tripotent $S = \tau\pi : D_1 \to D_1$. Then $S^2 = \pi$, $S^3 = S$, $S|_{V_0} = 0$, and $S|_E = \tau$. Direct evaluation of a multiple of $(Y-1)^e$ shows

$$\mathcal{A}(V_0) = \mathcal{X}(V_0) = 0, \qquad \mathcal{A} = \mathcal{A}\pi, \qquad \mathcal{X} = \mathcal{X}\pi.$$

Write $K_{\mathcal{A}} = \ker \mathcal{A}$ and $K_{\mathcal{A}\mathcal{X}} = \ker \mathcal{A} \cap \ker \mathcal{X}$.

Definition 4.1 (First integral obstruction)**.** The depth-three obstruction used in the Smith assembly is

$$\varepsilon_1(p,3) := \operatorname{rank}_k \begin{bmatrix} \mathcal{A} \\ \mathcal{A}S \end{bmatrix} - \operatorname{rank}_k \mathcal{A} = \operatorname{rank}_k\big((\mathcal{A}S)|_{K_{\mathcal{A}}}\big). \tag{4.2}$$

The second equality follows by projecting the stacked image onto its first coordinate.

Let

$$\jmath_{\mathrm{nu}} : E_{\mathcal{X}} \hookrightarrow E_{\mathcal{A}}, \qquad (\jmath_{\mathrm{nu}} f)_t = \begin{cases} f_u, & t = pu \text{ with } u \in \mathbb{Z}/N_0\mathbb{Z}, \\ 0, & p \nmid t. \end{cases} \tag{4.3}$$

Every nonunit class $t \bmod N$ has a unique expression $t = pu$ with $u \bmod N_0$, so this is a well-defined injective k-linear map. The notation $\jmath_{\mathrm{nu}}^{-1}$ below means its inverse only on the supported subspace $\operatorname{im} \jmath_{\mathrm{nu}}$.

From this point through section 6 assume that p is odd. The definition in definition 4.1 and the characteristic-free tripotent rank identity remain valid at $p = 2$, but the eigenspace splitting and the factor $1/2$ below do not.

Theorem 4.2 (Common-image reduction)**.** *Let $E = \operatorname{im}\pi = E_+ \oplus E_-$ be the ± 1 eigenspace decomposition of τ, and put*

$$\mathcal{A}_+ = \mathcal{A}|_{E_+}, \qquad \mathcal{A}_- = \mathcal{A}|_{E_-}, \qquad J_p^{\mathrm{full}} = \operatorname{im}\mathcal{A}_+ \cap \operatorname{im}\mathcal{A}_-.$$

These images and their intersection are k-subspaces of $E_{\mathcal{A}}$; no $\overline{\mathcal{O}}$-module structure is used here. Define

$$\begin{aligned} \mathrm{Fib}_p &= \ker[\mathcal{A}_+, -\mathcal{A}_-] \subset E_+ \oplus E_-, \\ N_{\mathrm{Fib}} &= \ker\mathcal{A}_+ \oplus \ker\mathcal{A}_-, \\ c_{\mathrm{Fib}} : \mathrm{Fib}_p / N_{\mathrm{Fib}} &\longrightarrow J_p^{\mathrm{full}}, \qquad\qquad [(x,y)] \longmapsto \mathcal{A}_+x = \mathcal{A}_-y. \end{aligned}$$

Then c_{Fib} is an isomorphism and

$$\varepsilon_1(p,3) = \dim_k J_p^{\mathrm{full}}. \tag{4.4}$$

Moreover, J_p^{full} is supported on the nonunit slopes and the transport $2\jmath_{\mathrm{nu}}^{-1}$ identifies it with

$$J_p^{\mathrm{red}} := 2\jmath_{\mathrm{nu}}^{-1}(J_p^{\mathrm{full}}) = \mathcal{X}(K_{\mathcal{A}}) \subset E_{\mathcal{X}}. \tag{4.5}$$

Proof. For any tripotent S, projection of $v \mapsto (\mathcal{A}v, \mathcal{A}Sv)$ onto its first coordinate shows that the rank gap is $\operatorname{rank}(\mathcal{A}S|_{\ker\mathcal{A}})$. Since $\mathcal{A}(V_0) = 0$, the zero sector disappears. On $E_+ \oplus E_-$ the invertible target change $(a,b) \mapsto ((a+b)/2, (a-b)/2)$ turns the stacked map into the direct sum of the two sector maps. Subtracting the rank of $\mathcal{A}$ leaves precisely the dimension of their common image. The map c_{Fib} is onto by the definition of the intersection, and its kernel before quotienting is exactly N_{Fib}.

Put $P_+ = (\pi + S)/2$ and $P_- = (\pi - S)/2$. The map

$$\Gamma : K_{\mathcal{A}} \longrightarrow \mathrm{Fib}_p, \qquad b \longmapsto (P_+b, -P_-b)$$

is onto: a preimage of $(x,y) \in \mathrm{Fib}_p$ is $b = x - y$. It has kernel V_0, and

$$c_{\mathrm{Fib}}[\Gamma(b)] = \frac{1}{2}\mathcal{A}Sb. \tag{4.6}$$

For $b \in K_{\mathcal{A}}$, the exact exchange identity is

$$(\mathcal{A}Sb)_t = 0 \quad (p \nmid t), \qquad (\mathcal{A}Sb)_{pu} = \mathcal{X}(b)_u.$$

Consequently

$$c_{\mathrm{Fib}}\Gamma = \frac{1}{2}\jmath_{\mathrm{nu}}\mathcal{X}|_{K_{\mathcal{A}}}. \tag{4.7}$$

Surjectivity of $c_{\mathrm{Fib}}\Gamma$ proves the support assertion and (4.5). The factor 2 is a unit because p is odd. $\square$

The reduced image has a sparse q-Pascal presentation. For $0 \le i < N$ and $0 \le k < e$, define

$$E_p(i,k)_r = (1-\zeta^{pr})^k C_i(r), \qquad C_i(r) = \sum_{j=i}^{N-1} \zeta^j \begin{bmatrix} j \\ i \end{bmatrix}_{\zeta^{-pr}}, \quad 0 \le r < p^2. \tag{4.8}$$

Let

$$\beta_p(n) = \sum_{d=1}^{n} p^{v_p(d)}, \qquad \lambda_i = \beta_p(N-1-i). \tag{4.9}$$

The source labels satisfy $\max(e - \lambda_i, 0) \le k < e$.

Corollary 4.3 (q-Pascal presentation). *After removing only columns proved identically zero,*

$$J_p^{\mathrm{red}} = \mathrm{im}(E_p), \qquad \varepsilon_1(p,3) = \mathrm{rank}_k(E_p). \tag{4.10}$$

Proof. The reduced fibre product $\mathrm{Fib}_p / N_{\mathrm{Fib}}$ is J_p^{full} by theorem 4.2. The Newton basis of $K_{\mathcal{A}}$ supplies the columns (4.8), and the exchange identity in the proof of theorem 4.2 identifies their image with the transported fibre product. Quotienting by the column-relation space proves (4.10). □

5 Exact $\mathfrak{q}$-Lucas block product and support

For $r \in \mathbb{Z}/p^2\mathbb{Z}$, put $Q_r = \zeta^{-pr}$ and let

$$L = L_r = \mathrm{ord}(Q_r) = \begin{cases} 1, & r = 0, \\ p, & r \neq 0,\ p \mid r, \\ p^2, & p \nmid r, \end{cases} \qquad M = N/L.$$

Write $i = dL + c$ with $0 \leq c < L$.

Lemma 5.1 ($\mathfrak{q}$-Lucas evaluation). *The evaluation $\mathfrak{q} \mapsto Q_r$ kills p and $\Phi_L(\mathfrak{q})$. Hence, for $0 \leq b < L$,*

$$\begin{bmatrix} aL + b \\ dL + c \end{bmatrix}_{Q_r} = \binom{a}{d} \begin{bmatrix} b \\ c \end{bmatrix}_{Q_r}. \tag{5.1}$$

Proof. If $L = p^a$, then in characteristic p, $\Phi_{p^a}(Z) = (Z-1)^{p^{a-1}(p-1)}$. For $L = p^2$ the order of $Q_r - 1$ is p, while for $L = p$ it is p^2; in both cases the displayed power has order e and vanishes in $\overline{\mathcal{O}}$. Sagan's congruence [17, Theorem 2.2] therefore applies. The case $L = 1$ is ordinary Lucas. □

Two finite identities will be used:

$$\sum_{a=d}^{M-1} \binom{a}{d} x^a = x^d (1-x)^{M-1-d}, \tag{5.2}$$

$$\sum_{b=c}^{L-1} x^b \begin{bmatrix} b \\ c \end{bmatrix}_Q = x^c \prod_{h=c+1}^{L-1} (1 - xQ^h), \tag{5.3}$$

where M is a power of p and $\Phi_L(Q) = 0$. The first follows by extracting a coefficient from a finite geometric series and using Frobenius. The second is the finite $\mathfrak{q}$-binomial theorem after the standard root-of-unity symmetry $\left[\begin{smallmatrix} L-1-c \\ t \end{smallmatrix}\right]_Q = (-1)^t Q^{-ct-t(t+1)/2} \left[\begin{smallmatrix} c+t \\ c \end{smallmatrix}\right]_Q$.

Theorem 5.2 (Exact block product and support). *For every odd prime, slope and index,*

$$C_i(r) = \zeta^i (1 - \zeta^L)^{M-1-d} \prod_{h=c+1}^{L-1} (1 - \zeta^{1-prh}). \tag{5.4}$$

Consequently

$$C_i(r) = \begin{cases} (-1)^i s^{N-1-i} + O(s^{N-i}), & N-1-i < e, \\ 0, & N-1-i \geq e, \end{cases} \tag{5.5}$$

and the column (i,k) in (4.8) is nonzero exactly when

$$\max(e-\lambda_i,0) \le k < e, \qquad N-1-i+pk < e. \tag{5.6}$$

Proof. Write $j = aL + b$ in (4.8). Applying lemma 5.1 separates the sum into the two factors in (5.2) and (5.3), which gives (5.4). Since $1-\zeta^L = -s^L$ and every $1-\zeta^{1-prh}$ begins with $-s$, its total order is

$$L(M-1-d) + (L-1-c) = N-1-i.$$

The parity of the number of negative factors is the parity of i, proving (5.5). On a unit slope, $1-\zeta^{pr} = -r_0 s^p + O(s^{2p})$; this proves sufficiency in (5.6). Nonunit slopes have no smaller prefactor order, proving necessity. □

The equality (5.4) belongs to the truncated target $\overline{\mathcal{O}}$; it is not an equality in the unquotiented group algebra. It resolves cancellation inside a single column but does not yet control relations among columns.

6 Boundary generation and the Newton-minors theorem

Put $u = \zeta^p = 1 + s^p$ and $T_r = 1 - u^r$. Reverse the column index by

$$\mathcal{C}_t(r) = C_{N-1-t}(r), \qquad F_{t,K}(r) = T_r^K \mathcal{C}_t(r). \tag{6.1}$$

The surviving labels are exactly

$$K + \beta_p(t) \ge e, \qquad t + pK < e. \tag{6.2}$$

Let $W = \mathrm{im}(E_p) \subset \overline{\mathcal{O}}^{p^2}$, initially only as a k-space.

Lemma 6.1 (Scalar recurrence and module closure)**.** *With $c = s\zeta^{-1}$,*

$$cF_{t,K} = \sum_{\ell=1}^{t+1} (-1)^{\ell+1} \binom{t+1}{\ell} F_{t,K+\ell} - F_{t+1,K}. \tag{6.3}$$

All coefficients are in k and independent of r. Consequently W is an $\overline{\mathcal{O}}$-submodule of $\overline{\mathcal{O}}^{p^2}$.

Proof. The exact recurrence before expansion is $F_{t+1,K} = -cF_{t,K} + [t+1]_{u^r} F_{t,K+1}$. Since

$$T_r[t+1]_{u^r} = 1 - (1-T_r)^{t+1},$$

binomial expansion gives (6.3). Every nonzero term remains admitted or is zero by the support boundary, so $cW \subseteq W$. Finally $c^e = 0$ and $s = c/(1-c) = c + \cdots + c^{e-1}$, hence $sW \subseteq W$. □

Define

$$\mu_p(x) = \min\{t \ge 0 : \beta_p(t) \ge x\}, \qquad t_K = \mu_p(e-K), \qquad G_K = F_{t_K,K}, \tag{6.4}$$

and

$$a_K = t_K + pK, \qquad H_p = \#\{0 \le K < p(p-1) : a_K < e\}. \tag{6.5}$$

Here G_K is a vector and H_p is an integer. These types are fixed throughout the paper.

Lemma 6.2 (Boundary generation)**.** *The nonzero boundary indices are $0 \le K < H_p$, and*

$$W = \sum_{K=0}^{H_p-1} \overline{\mathcal{O}} G_K. \tag{6.6}$$

Proof. Since $\mu_p(x) - \mu_p(x-1) \in \{0,1\}$, $a_{K+1} - a_K \ge p-1$, so the nonzero indices form an initial interval. For the induction in t, one needs the following Lucas admission step: if $\binom{t}{\ell} \not\equiv 0 \pmod{p}$, then

$$\ell \ge p^{v_p(t)} = \beta_p(t) - \beta_p(t-1).$$

Indeed, Lucas forces the first $v_p(t)$ digits of ℓ to vanish. Apply (6.3) at $(t-1, K)$ and solve for $F_{t,K}$. Every term with nonzero coefficient remains admitted by the displayed inequality and has smaller first index; induction reaches G_K. $\square$

Lift to $R = k[[s]]$. Define

$$\widetilde{\mathcal{C}}_t(r) = \zeta^{-1} \prod_{\ell=1}^{t} (\zeta^{-1} - u^{r\ell}), \qquad \widetilde{G}_K(r) = T_r^K \widetilde{\mathcal{C}}_{t_K}(r). \tag{6.7}$$

Put $X_r = [r]_u$ and $Q_j(r) = \begin{bmatrix} r \\ j \end{bmatrix}_u$. The matrix $(Q_j(r))_{0 \le r,j < p^2}$ is lower unitriangular, so the Q_j form an R-basis of R^{p^2}.

Lemma 6.3 (Coefficient cost)**.** *There are a unit $\eta_t(s)$ and coefficients $h_{t,m}(s)$ such that*

$$\widetilde{\mathcal{C}}_t = s^t \eta_t(s) \sum_{m \ge 0} h_{t,m}(s) X^m, \qquad h_{t,0} = 1, \qquad v_s(h_{t,m}) \ge (p-1)m. \tag{6.8}$$

Proof. Each factor in (6.7) is

$$-c + \sum_{d=1}^{\ell} (-1)^{d+1} \binom{\ell}{d} (1-u)^d X^d.$$

After the constant term is factored, a selection with total X-degree m has extra s-order at least pm minus the number of selected factors, hence at least $(p-1)m$. The all-constant selection is unique. $\square$

Carlitz's integral identity [3] is

$$X^n = \sum_{j=0}^{n} u^{\binom{j}{2}} S_u[n,j] [j]_u! \, Q_j. \tag{6.9}$$

For $j < p^2$, put

$$\phi_j = v_s([j]_u!) = p(\beta_p(j) - j), \qquad \tau_p(K) = a_K + \phi_K = \mu_p(e-K) + p\beta_p(K). \tag{6.10}$$

The sequence $\tau_p(K)$ is strictly increasing.

Theorem 6.4 (Newton-minors theorem)**.** *Let $\mathcal{M}_p$ be the $p^2 \times H_p$ matrix of the lifted boundary columns in the $\mathfrak{q}$-Newton row basis. If I_r is its ideal of $r \times r$ minors, then*

$$v_s(I_r) = \sum_{K=0}^{r-1} \tau_p(K) \qquad (1 \le r \le H_p). \tag{6.11}$$

Thus the Smith exponents over the DVR R are exactly $\tau_p(0), \ldots, \tau_p(H_p - 1)$.

Proof. Combining lemma 6.3 with (6.9), every path $K \to K+m \to j$ contributing to row j of column K has order at least $a_K + \phi_j$; equality occurs for $m = 0, j = K$. For column indices $K_1 < \cdots < K_r$ and row indices $J_1 < \cdots < J_r$, every determinant monomial therefore has order at least

$$\sum_i a_{K_i} + \sum_i \phi_{J_i} \geq \sum_{K=0}^{r-1} \tau_p(K).$$

In the principal minor on rows and columns $0, \ldots, r-1$, the identity permutation with paths $m = 0, j = K$ attains this bound uniquely. A path with $m > 0$ has positive extra cost; with $m = 0$, Carlitz support requires $\pi(K) \leq K$ for each column, forcing the identity permutation. Hence the minimum term cannot cancel, proving (6.11). Successive differences of determinantal-ideal valuations are the DVR Smith exponents. □

7 Arithmetic evaluation and the filtered common image

Reduction of the DVR Smith form modulo s^e, together with lemma 6.2, gives the obstruction dimension

$$\varepsilon_1(p,3) = \sum_{K=0}^{p(p-1)-1} \left[e - \mu_p(e-K) - p\beta_p(K)\right]_+ . \tag{7.1}$$

Put

$$L_K = e - \tau_p(K) = e - \mu_p(e-K) - p\beta_p(K). \tag{7.2}$$

Lemma 7.1 (Exact inversion of β_p)**.** *Let $p > 3$, write $p = 6h + \rho$ with $\rho \in \{1, 5\}$, and set $d = 2p - 1$. For $0 \leq K \leq M$ choose the branch constants*

$$\begin{array}{c|c|c|c|c} \rho & A & B_0 & c & M \\ \hline 1 & 2h & 2h & 4h & h(2p+1) \\ 5 & 2h+1 & 5h+4 & 7h+5 & \dfrac{p^2-1}{3} \end{array} \tag{7.3}$$

and define

$$\begin{aligned} R_K &= e - K - Ap(3p-2), \\ B_K &= \left\lceil \frac{R_K - (p-1)}{d} \right\rceil = B_0 - \left\lfloor \frac{K+c}{d} \right\rfloor, \\ C_K &= \max(0, R_K - dB_K), \\ \delta_K &= \max(0, dB_K - R_K) = \max\big(0, ((K+c) \bmod d) - (p-1)\big). \end{aligned} \tag{7.4}$$

Then A, B_K, C_K are the base-p digits of the least integer whose β_p-value is at least $e-K$, and

$$\mu_p(e-K) = Ap^2 + R_K - (p-1)B_K + \delta_K. \tag{7.5}$$

Consequently

$$\boxed{L_K = (p-1)\left(2Ap + B_0 - K - p\left\lfloor \frac{K}{p} \right\rfloor - \left\lfloor \frac{K+c}{2p-1} \right\rfloor\right) - \delta_K.} \tag{7.6}$$

Proof. For $0 \le t < p^3$ the valuation-count identity is

$$\beta_p(t) = t + (p-1)\left\lfloor \frac{t}{p} \right\rfloor + (p^2 - p)\left\lfloor \frac{t}{p^2} \right\rfloor.$$

Writing $t = Ap^2 + Bp + C$ with $0 \le A, B, C < p$ therefore gives

$$\beta_p(t) = Ap(3p-2) + B(2p-1) + C. \tag{7.7}$$

For the intervals in (7.3), direct substitution at $K = 0, M$ keeps the first digit equal to the displayed A. Once A is fixed, the least B for which the residual target can be reached with $0 \le C < p$ is B_K in (7.4), and the least last digit is C_K. Since $C_K = R_K - dB_K + \delta_K$, substitution into $Ap^2 + B_K p + C_K$ gives (7.5). The identity $R_0 - (p-1) = B_0 d - c$ gives the second formula for B_K and the residue formula for δ_K. Finally, for $K < p^2$, $\beta_p(K) = K + (p-1)\lfloor K/p \rfloor$; inserting this and (7.5) into (7.2) gives (7.6). □

Lemma 7.2 (Positive-tail endpoints and residue sums)**.** *With the branch constants of lemma 7.1,*

$$L_{M-1} = p - 1, \qquad L_M = 0, \qquad L_K > 0 \Longleftrightarrow 0 \le K < M. \tag{7.8}$$

Moreover, the two floor sums and the periodic ramp in (7.6) are

$$\begin{array}{c|c|c}
 & p = 6h+1 & p = 6h+5 \\
\hline
\displaystyle\sum_{K=0}^{M-1} \left\lfloor \frac{K}{p} \right\rfloor & ph(2h-1) + 2h^2 & ph(2h+1) + (2h+1)(4h+3) \\
\displaystyle\sum_{K=0}^{M-1} \left\lfloor \frac{K+c}{2p-1} \right\rfloor & (2p-1)\dfrac{h(h-1)}{2} + 6h^2 & (2p-1)\dfrac{h(h+1)}{2} + (h+1)(p-1) \\
\displaystyle\sum_{K=0}^{M-1} \delta_K & 3h^2 p & (h+1)\dfrac{p(p-1)-h}{2}.
\end{array} \tag{7.9}$$

Proof. Substitution of (7.3) into (7.6) gives the two endpoint equalities. Strict increase of $\tau_p(K)$ from theorem 6.4 makes L_K strictly decreasing, so they prove the last assertion of (7.8). For the first row of (7.9), write $M = ap + b$ and sum a complete residue blocks. For the second and third rows, write $K + c = j(2p-1) + r$ and sum separately over $0 \le r < 2p - 1$ and the terminal block; the ramp is nonzero precisely for $p \le r < 2p - 1$. The complete residue decompositions are displayed in section B; they give (7.9) without an asymptotic or interpolation step. □

Theorem 7.3 (Closed odd-prime obstruction)**.** *For odd primes,*

$$\varepsilon_1(p,3) = \begin{cases} 18, & p = 3, \\ \frac{(p-1)^2(4p^3+5p^2+2p+4)}{36}, & p > 3,\ p \equiv 1 \pmod 6, \\ \frac{p(p+1)(8p^3-18p^2+11p-5)}{72}, & p \equiv 5 \pmod 6. \end{cases} \tag{7.10}$$

Proof. For $p > 3$, lemma 7.2 reduces (7.1) to $\sum_{K=0}^{M-1} L_K$. Summing (7.6), using $\sum_{K=0}^{M-1} K = M(M-1)/2$, and substituting the three exact rows of (7.9) gives the two polynomials in (7.10). For $p = 3$, the active thresholds are $9, 12, 15$ and $e = 18$, so the sum is $(18-9) + (18-12) + (18-15) = 18$. □

Theorem 7.4 (Filtered common-image profile)**.** *As an $\overline{\mathcal{O}}$-module,*

$$W \cong \bigoplus_{\substack{0\le K<H_p\\ \tau_p(K)<e}} \overline{\mathcal{O}}/(s^{e-\tau_p(K)}). \tag{7.11}$$

For the ambient filtration $F^d_{\mathrm{amb}}W = W \cap s^d\overline{\mathcal{O}}^{p^2}$,

$$\dim_k(F^d_{\mathrm{amb}}W/F^{d+1}_{\mathrm{amb}}W) = \#\{0 \le K < H_p : \tau_p(K) \le d < e\}. \tag{7.12}$$

Hence

$$\mathrm{Hilb}^{\mathrm{amb}}_W(z) = \sum_{\substack{0\le K<H_p\\ \tau_p(K)<e}} z^{\tau_p(K)}\frac{1-z^{e-\tau_p(K)}}{1-z}. \tag{7.13}$$

The minimal number of generators is

$$\mathrm{ngen}_{\overline{\mathcal{O}}}(W) = \dim_k(W/sW) = \begin{cases} 3, & p=3,\\ \dfrac{(p-1)(2p+1)}{6}, & p\equiv 1 \pmod 6,\\ \dfrac{p^2-1}{3}, & p \equiv 5 \pmod 6. \end{cases} \tag{7.14}$$

Proof. Invertible Smith row operations over R preserve every ambient power $s^dR^{p^2}$. Reduction modulo s^e turns the active diagonal with threshold $\tau_p(K)$ into the embedded ideal $s^{\tau_p(K)}\overline{\mathcal{O}}$, proving (7.11)–(7.12). Summing the successive dimensions gives (7.13). Nakayama's lemma counts one generator for each nonzero cyclic summand, and the positive-tail counts in the proof of theorem 7.3 give (7.14). □

The boundary-column count and generator count are different invariants:

p	3	5	7	11
H_p	3	12	28	71
$\mathrm{ngen}_{\overline{\mathcal{O}}}(W)$	3	8	15	40.

(7.15)

Corollary 7.5 (Truncation redundancy)**.**

$$H_p - \mathrm{ngen}_{\overline{\mathcal{O}}}(W) = \#\{0 \le K < H_p : \tau_p(K) \ge e\} \tag{7.16}$$

counts the Smith directions in the boundary presentation eliminated by truncation modulo s^e.

This is a numerical structural statement. It does not assert that a particular named subset of the original boundary columns is zero.

8 The cross-chart transfer defect

Let $\eta_2 : \mathsf{Q}_2/p\mathsf{Q}_2 \to p\mathsf{Q}_3/p^2\mathsf{Q}_3$ be induced by transfer and put $d_p = \dim_k \ker \eta_2$. The equality

$$\tau_3(\mathsf{Q}_2) \cap p^2\mathsf{Q}_3 = \tau_3(p\mathsf{Q}_2) \tag{O2}$$

is equivalent to $d_p = 0$.

Lemma 8.1 (Defect bookkeeping)**.** *Let* $T_{\geq 3} = m_3 + m_4 + m_5$*. Then*

$$T_{\geq 3} = a_0 + \varepsilon_1(p, 3) + d_p. \tag{8.1}$$

Proof. The multiscale identity gives $p^3\mathsf{Q}_3 = \tau_3(p\mathsf{Q}_2)$. The natural map $p^2\mathsf{Q}_3/p^3\mathsf{Q}_3 \to p^2(\mathsf{Q}_3/\tau_3\mathsf{Q}_2)$ is onto, with kernel

$$\frac{\tau_3\mathsf{Q}_2 \cap p^2\mathsf{Q}_3}{\tau_3(p\mathsf{Q}_2)},$$

whose dimension is d_p. The relative top-shell calculation gives $a_0 + \varepsilon_1$ for the target dimension, proving (8.1). □

We now define the carry objects used to kill this defect. Let

$$\mathscr{R}_2 = k[G, H]/(G^{N_0} - 1, H^{N_0} - 1), \qquad x = G - 1, \quad y = H - 1,$$

identified with the k-space of functions on the coarse point set $G_2 = (\mathbb{Z}/N_0\mathbb{Z})^2$. With $\iota([g]) = [-g]$, use the Frobenius pairing $\langle a, b\rangle = [1](a\iota(b))$. For $\delta \in \mathbb{P}^1(k)$ choose

$$\begin{array}{c|cc} \delta & g_\delta & t_\delta \\ \hline \delta \in k & GH^\delta & H \\ \infty & H & G, \end{array} \qquad z_\delta = g_\delta - 1, \quad v_\delta = t_\delta - 1,$$

and define the tangent-bundle ideal and line code

$$I_\delta = (z_\delta, v_\delta^p)^p, \qquad \mathcal{C}_{2,\delta} = I_\delta^\perp, \qquad \mathcal{C}_2 = \sum_{\delta \in \mathbb{P}^1(k)} \mathcal{C}_{2,\delta}. \tag{8.2}$$

The normal-to-tangent conversion is

$$\kappa(0) = \infty, \qquad \kappa(d) = -d^{-1} \ (d \in k^\times), \qquad \kappa(\text{cross}) = 0. \tag{8.3}$$

For $b \in K_{\mathcal{A}}$ choose a compatible integral lift $\widetilde{b}$ in the fixed cyclotomic lattice and write its first-chart remainders as

$$A_t(\widetilde{b}) = \sum_{j<e} a_{t,j}\zeta^j.$$

Every $a_{t,j}$ is divisible by p. If moreover $b \in K_{\mathcal{AX}}$, write

$$X_u(\widetilde{b}) = \sum_{j<e} x_{u,j}\zeta^j;$$

then every $x_{u,j}$ is divisible by p as well. If $o \in \mathbb{Z}/N\mathbb{Z}$, let $[c_\bullet]_o^{<e}$ mean $c_{[o]_N}$ when $0 \leq [o]_N < e$ and zero when $e \leq [o]_N < N$. For $(r, s) \in G_2$ put

$$o_t(r, s) = [e + r + t(e + s)]_N, \qquad o_u^{\mathcal{X}}(r, s) = [pu(e + r) + e + s]_N. \tag{8.4}$$

The first-chart residue carries and the cross carry are the typed maps

$$\begin{gathered} \mathcal{H}_d : K_{\mathcal{A}} \longrightarrow \mathscr{R}_2 \quad (d \in k), \\ \mathcal{H}_X : K_{\mathcal{AX}} \longrightarrow \mathscr{R}_2, \\ \mathcal{H}_A := \sum_{d \in k} \mathcal{H}_d : K_{\mathcal{A}} \longrightarrow \mathscr{R}_2. \end{gathered}$$

Their values on compatible lifts are

$$\begin{aligned}\mathcal{H}_d(b)(r,s) &= \sum_{\substack{t \bmod N \\ t\equiv d \bmod p}} \frac{[a_{t,\bullet}]^{<e}_{o_t(r,s)}}{p} \pmod p,\\ \mathcal{H}_X(b)(r,s) &= \sum_{u \bmod N_0} \frac{[x_{u,\bullet}]^{<e}_{o^{\mathcal{X}}_u(r,s)}}{p} \pmod p.\end{aligned} \tag{8.5}$$

Division is performed only after an integral candidate relation has been combined, so its coefficients are divisible by p. If the compatible lift is changed by ph, the change in (8.5) is an ordinary carrier in the corresponding line bundle; hence the map

$$\delta_1 : K_{\mathcal{A}\mathcal{X}} \longrightarrow \mathscr{R}_2/\mathcal{C}_2, \qquad \delta_1(b) = [\mathcal{H}_A(b) + \mathcal{H}_X(b)] \tag{8.6}$$

is independent of the lift.

For an integral representative define the one-way exchange

$$\widetilde{S}(\widetilde{b}) = \mathrm{Rem}_{\Phi_{p^3}(X)}\, \widetilde{b}(Y,X), \tag{8.7}$$

with monic remainder in X. Modulo p, $\Phi_{p^3}(X) = (X-1)^e$; hence this reduction is exactly the tripotent $S = \tau\pi : D_1 \to D_1$ defined in section 4, rather than a second operator with the same name.

Lemma 8.2 (Carry transposition)**.** *For odd p,*

$$S(K_{\mathcal{A}\mathcal{X}}) \subset K_{\mathcal{A}}, \qquad \mathcal{H}_0(Sb)(r,s) = \mathcal{H}_X(b)(s,r). \tag{8.8}$$

If P is point transposition, $(Pf)(r,s) = f(s,r)$, then

$$P(\mathcal{C}_{2,\infty}) = \mathcal{C}_{2,0}. \tag{8.9}$$

Proof. Monic division in (8.7) gives the integral identities

$$A_t(\widetilde{S\widetilde{b}}) = \widetilde{b}(\zeta^t,\zeta), \qquad A_{pu}(\widetilde{S\widetilde{b}}) = X_u(\widetilde{b}). \tag{8.10}$$

For a unit t, the automorphism $\zeta \mapsto \zeta^t$ transports the vanishing of $\mathcal{A}(b)$ at t^{-1} to the first expression in (8.10); for a nonunit $t = pu$, the second kernel condition applies. This proves the inclusion. The second identity in (8.10) identifies the integer coefficient arrays, while

$$o_{pu}(r,s) = e + r + pu(e+s) = o^{\mathcal{X}}_u(s,r) \pmod N.$$

Thus clipping, division and summation agree term by term, proving (8.8). Finally, transposition takes the p depth-two lines with normals $(1,pu)$ to those with normals $(pu,1)$, proving (8.9). ☐

To state the seed calculation, let $\rho_p : D_1 \to \mathscr{R}_2$ reduce both exponents modulo N_0, put

$$P_j(X,Y) = \prod_{a=0}^{j-1}(Y - X^a), \qquad b_{j,\nu} = (X-1)^\nu P_j(X,Y), \qquad \nu \geq \max(e - \beta_p(j), 0), \tag{8.11}$$

and let Π_0 retain the monomials whose Y-degree is divisible by p.

Lemma 8.3 (Newton kernel basis)**.** *The vectors*

$$b_{j,\nu}, \qquad 0 \le j < N, \quad \max(e - \beta_p(j), 0) \le \nu < e,$$

form a k-basis of $K_{\mathcal{A}}$.

Proof. Each P_j is monic of degree j in Y, so the transition from $\{(X-1)^\nu Y^j\}$ to $\{(X-1)^\nu P_j\}$ is unitriangular over $k[X]/((X-1)^e)$. Moreover

$$P_j(\zeta, \zeta^t) = P_j(\zeta, \zeta^j) \begin{bmatrix} t \\ j \end{bmatrix}_\zeta,$$

where the Gaussian-Pascal matrix is lower unitriangular in t, j. Invertible row operations therefore diagonalize first-chart evaluation with diagonal entries

$$d_j = P_j(\zeta, \zeta^j) = \prod_{a=0}^{j-1} (\zeta^j - \zeta^a).$$

For $1 \le d \le j$, the factor $\zeta^d - 1$ is a unit times $s^{p^{v_p(d)}}$; hence d_j is a unit times $s^{\beta_p(j)}$. In $\overline{\mathcal{O}} = k[s]/(s^e)$ the kernel of multiplication by d_j is exactly $s^{\max(e-\beta_p(j),0)}\overline{\mathcal{O}}$, whose displayed powers of s form a k-basis. Taking the direct sum over j proves the claim. $\square$

For a first-chart residue d, the $p+1$ mixed differences generating $I_{\kappa(d)}$ give functionals $\lambda_{d,c}(b)$, $0 \le c \le p$, by applying $z_{\kappa(d)}^{p-c}(v_{\kappa(d)}^p)^c$ to $\mathcal{H}_d(b)$.

Lemma 8.4 (Seed-defect factorization and annihilation)**.** *For every odd prime, every residue d and $b \in K_{\mathcal{A}}$,*

$$\lambda_{d,c}(b) = \rho_p(b) S_{d,c}, \qquad 0 \le c \le p, \tag{8.12}$$

where, with $y_d = G^{-d}H - 1$,

$$\begin{aligned} S_{d,0} &= -y_d^p + x^{p(p-1)} y_d^{p(p-1)} C_p(y_d), \\ S_{d,c} &= x^{pc}(y_d^{N_0-c} - y_d^{p-c}) \quad (1 \le c < p), \\ S_{d,p} &= 0, \qquad C_p(Y) = \sum_{a=1}^{p-1} \left(\frac{1}{p}\binom{p}{a} \bmod p \right) Y^a. \end{aligned} \tag{8.13}$$

The auxiliary carrier seeds are

$$\begin{aligned} B^0_{d,0} &= x^{p(p-1)} y_d^{p(p-1)} C_p(y_d), \\ B^0_{d,c} &= x^{pc} y_d^{N_0-c} \quad (1 \le c < p), \quad B^0_{d,p} = 0. \end{aligned}$$

They satisfy, on every admissible Newton basis vector,

$$\rho_p(b_{j,\nu}) B^0_{d,c} = 0, \qquad \rho_p(\Pi_0 b_{j,\nu}) B^0_{d,c} = 0. \tag{8.14}$$

The second equality is the Cartier constant-term assertion.

Proof. Let $\sigma(n, o)$ be the coefficient of ζ^o in the standard representative of ζ^n modulo Φ_{p^3}, extended by zero for $e \le o < N$. The last cyclotomic block gives exactly

$$\sigma(n, o) - \mathbf{1}_{o \equiv n \ (N)} = -\mathbf{1}_{[n]_N \ge e} \mathbf{1}_{o \equiv n \ (N_0)}. \tag{8.15}$$

We give the residue-packet calculation explicitly. For a source monomial $m = X^iY^j$, put $J = [j]_p$ and, at the point (r, s), put $S = [s]_p$. Its undivided residue-d carrier is

$$T_d(m)(r, s) = \sum_{u \bmod N_0} \sigma\big(i + (d + pu)j, o_{d+pu}(r, s)\big). \tag{8.15a}$$

Write $u = u_0 + pv$, with $u_0, v \in k$. The common N_0-tail condition is

$$r + (d + pu_0)s \equiv i + (d + pu_0)j \pmod{N_0};$$

denote its solution set in u_0 by U. For $u_0 \in U$, the source and clipped-point top-block digits have the form $A + Jv$ and $B + Sv$. By (8.15), the complete v-packet is

$$\mathsf{P}(A, B; J, S) = \sum_{v\in k} \mathbf{1}_{B+Sv\neq -1}\big(\mathbf{1}_{A+Jv=B+Sv} - \mathbf{1}_{A+Jv=-1}\big). \tag{8.15b}$$

The four possibilities are as follows; the last column is evaluated for each $u_0 \in U$.

(J, S)	$\lvert U\rvert$	$\mathsf{P}(A, B; J, S) \pmod p$
$J \neq S,\ J \neq 0$	0 or 1	0
$J \neq S,\ J = 0$	0 or 1	1
$J = S \neq 0$	0 or p	$-1 \quad (\sum_{u_0\in U} \mathsf{P} = 0)$
$J = S = 0$	0 or p	$p\,\mathbf{1}_{B\neq -1}(\mathbf{1}_{A=B} - \mathbf{1}_{A=-1}) = 0$

Indeed, in the first row the equations $A + Jv = B + Sv$ and $A + Jv = -1$ each have one solution, and either both are excluded or neither is. In the second row the first equation contributes once; if $A = -1$, its excluded contribution is replaced by $-(p-1) = 1$. In the third row the packet is -1 whether $A = B$ or not, and the p values of u_0 cancel modulo p. The fourth row is constant in v and is visibly a multiple of p. Consequently, if

$$N_{d,u_0;i,j} = \{(r, s) : r + (d + pu_0)s = i + (d + pu_0)j \pmod{N_0}\},$$

then

$$T_d(m) \bmod p = \mathbf{1}_{J=0} \sum_{u_0\in k} \mathbf{1}_{N_{d,u_0;i,j}}. \tag{8.15c}$$

Thus the ordinary carrier is in the residue-d line bundle, including all seam and wall contributions.

To retain the information needed after division by p, let $L_d(m)$ be the integral 0–1 line sum in (8.15c) and write $T_d = L_d + pR_d$. With $\rho_p(m) = G^iH^j$, the same four rows before the final reduction give

$$T_d(m) - \rho_p(m)T_d(1) \equiv \begin{cases} -\rho_p(m)L_d(1), & J \neq 0, \\ -p \displaystyle\sum_{u_0\in W_m} \mathbf{1}_{N_{d,u_0;i,j}}, & J = 0, \end{cases} \pmod{p^2}, \tag{8.15d}$$

where W_m consists of the u_0 for which the constant source top-block digit is the last block. For clarity, at the translated point $([r - i]_{N_0}, [s - j]_{N_0})$ the seed has point digit $S - J$ and the same tail set U. Canonicalizing that point changes only the constant term of the affine top-block test in u_0, not its linear coefficient. If that coefficient is nonzero, both tests have one solution; if it is zero, each has zero or p solutions. Their counts therefore agree modulo p, and the complete v-packet supplies the second factor of p in (8.15d).

Let $D_{d,c} = z_{\kappa(d)}^{p-c}(v_{\kappa(d)}^p)^c$. It annihilates every line sum in (8.15c) modulo p. If $\mathcal{B}_{d,c}$ denotes the resulting line Bockstein, then on a combined first-chart relation

$$\lambda_{d,c} = \mathcal{B}_{d,c}(L_d) + D_{d,c}R_d.$$

Put $B^0_{d,c} = \mathcal{B}_{d,c}(L_d(1))$ and $S_{d,c} = \lambda_{d,c}(1)$. Applying $D_{d,c}$ to (8.15d), dividing only after the complete relation has been combined, and extending linearly gives

$$\lambda_{d,c}(b) - \rho_p(b)S_{d,c} = -\rho_p((1-\Pi_0)b)B^0_{d,c}. \tag{8.15e}$$

The exact values of $B^0_{d,c}$ and $S_{d,c}$ are obtained by putting $E = G^p$, $T = E - 1 = x^p$,

$$S_u(E) = \frac{E^u - 1}{E - 1}, \qquad h(E) = \frac{(E-1)^p - (E^p - 1)}{p} \pmod{p}.$$

In $k[T]/(T^p)$ one has $h(1+T) = -\log(1+T)$. For $A_n(T) = \sum_{u\in k} E^u S_u(E)^n$, finite-field moments give

$$hA_n = 0 \ (0 \le n \le p-2), \qquad hA_{p-1} = T,$$

and, for $0 \le m \le p-2$,

$$hA_{p+m} = \frac{(-1)^m}{m+1}T^{p-1-m} \pmod{T^{p-m}}. \tag{8.16}$$

Indeed, with $\phi(w) = \sum_{a=1}^{p-1}\binom{w}{a}T^{a-1}$, $\phi'(w) = -(h/T)(1+T)^w$ and $\phi(w)^p = w^p$ modulo T^p; summing $-T\phi(u)^n\phi'(u)$ over $u \in k$ leaves only positive exponents divisible by $p-1$. This proves (8.16), and substitution yields (8.13).

Every admissible label in (8.11) satisfies $j + \nu \ge N_0 - p$. Together with the displayed degrees of $B^0_{d,c}$, this proves the first equality in (8.14) by the nilpotence threshold in $k[x,y]/(x^{N_0}, y^{N_0})$; the shears below preserve this filtration.

For the Cartier term first take $d = 0$ and work in

$$\mathscr{A} = k[G,Z]/(G^{N_0} - 1, Z^p - 1), \qquad E = G^p, \quad T = E - 1 = x^p, \quad z = Z - 1.$$

The root set of $Q(U) = \prod_{b\in k}(U - E^b)$ is stable under multiplication by E, so $Q(EU) = Q(U)$. If q_i is an intermediate coefficient, then $(E^i - 1)q_i = 0$; because $(E^i-1)/T$ is a unit for $1 \le i < p$, each q_i is divisible by T^{p-1}. The constant coefficient is -1, and $Q(1) = 0$. Setting $U = Z$ and using $z^p = 0$ therefore gives

$$\prod_{b\in k}(Z - G^{pb}) \in x^{p(p-1)}z\mathscr{A}.$$

Among $a = 0, \dots, N_0 - p$ there remain $(p-1)^2$ indices not divisible by p, and each corresponding factor $Z - G^a$ lies in (x, z). A monomial surviving their product with the preceding ideal would have the form

$$x^{p(p-1)+\alpha}z^{1+\gamma}, \qquad \alpha \le p-1, \quad \gamma \le p-2, \quad \alpha + \gamma \ge (p-1)^2.$$

This is impossible for odd p, since $(p-1)^2 > 2p - 3$. Hence, with $j_0 = N_0 - p + 1$,

$$P_{j_0}(G,Z) = \prod_{a=0}^{N_0-p}(Z - G^a) = 0 \quad \text{in } \mathscr{A}, \tag{8.17}$$

and therefore $P_j(G, Z) = 0$ for every $j \geq j_0$.

Now $\rho_p(\Pi_0 b_{j,\nu})$ is x^ν times a polynomial in H^p. After multiplication by a nonzero $B^0_{0,c}$, every positive power of $H^p - 1 = y^p$ exceeds the y^{N_0-1} boundary; only evaluation at $H^p = 1$ can survive. Its coefficient is the image of $P_j(G, Z)$ modulo $Z^p - 1$, which vanishes for $j \geq j_0$ by (8.17). If $j < j_0$, then

$$\beta_p(j) \leq \beta_p(N_0 - p) = 2p^2 - 3p + 1, \qquad \nu \geq e - \beta_p(j) \geq (p-1)^3 \geq N_0 - p;$$

the x-power in every nonzero auxiliary seed makes the product zero. This proves the second equality of (8.14) for $d = 0$.

For general d, let $\Theta_d b(X, Y) = b(X, X^d Y)$ and let $\Phi_d(r, s) = (r + ds, s)$. The first map preserves $K_\mathcal{A}$ and $\Pi_0\Theta_d = \Theta_d\Pi_0$, while

$$\rho_p(\Theta_d b) = \Phi_d^{-1}(\rho_p(b)), \qquad B^0_{d,c} = \Phi_d(B^0_{0,c}).$$

Conjugating the $d = 0$ identity proves (8.14) for every residue. Substitution in the packet defect identity now gives (8.12). ☐

Lemma 8.5 (Carrier vanishing)**.** *For every prime $p \geq 5$ and every $d \in k$,*

$$\mathcal{H}_d(K_\mathcal{A}) \subset \mathcal{C}_{2,\kappa(d)}. \tag{8.18}$$

In particular, $\mathcal{H}_0(K_\mathcal{A}) \subset \mathcal{C}_{2,\infty}$ and $\mathcal{H}_A(K_\mathcal{A}) \subset \mathcal{C}_2$.

Proof. For $p \geq 5$, every label in (8.11) satisfies the sharper bound $j + \nu \geq 2p^2 - p - 1$. Since the least total degree of a nonzero seed in (8.13) is p, lemma 8.4 gives $\lambda_{d,c}(b_{j,\nu}) = 0$ for all c. The mixed differences generate $I_{\kappa(d)}$, so the Frobenius pairing in (8.2) identifies their joint kernel with $I^\perp_{\kappa(d)} = \mathcal{C}_{2,\kappa(d)}$. Linearity on the Newton basis proves (8.18). ☐

Lemma 8.6 (Second-domain cancellation)**.** *In the recursive top-shell bases write*

$$B_3^\mathsf{T} = \begin{pmatrix} B_2^\mathsf{T} & K \\ 0 & Z \end{pmatrix}, \qquad T = \begin{pmatrix} \mathcal{A} & 0 \\ \mathcal{X} & pV \end{pmatrix}.$$

For $f \in \mathcal{J}_2$, put $r_f = K^\mathsf{T} f$. Then $T^\mathsf{T} r_f$ is divisible by p, and, if

$$s_f = p^{-1} T^\mathsf{T} r_f \pmod{p} = (s_{f,1}, s_{f,2})$$

according to the two domain blocks of T, one has $s_{f,2} = 0$.

Proof. For $(a, b) \in k^2$, let $F_{a,b}$ be the indicator of the fibre of $G_2 \to k^2$ above (a, b) and put

$$Y_b = \sum_{a \in k} F_{a,b}, \qquad \mathsf{H}_b = Y_b - F_{0,b}, \qquad U = \sum_{b \neq 0} Y_b.$$

For a fine point z above (a, b), the sum of one line through z in each projective direction is $\mathbf{1} - F_{a,b}$ modulo p. Thus every $F_{a,b}$, and hence Y_b, H_b and U, belongs to the depth-two line code $\mathcal{C}_2$.

The coefficient sum for the first domain block is

$$\overline{KT_1}(X^i Y^j) = \mathbf{1}_{\{\bar{j}=0\}} U + \mathbf{1}_{\{\bar{i}=0\}} \mathsf{H}_{\bar{j}}.$$

Pairing with $f \in \mathcal{J}_2 = \mathcal{C}_2^\perp$ proves the required divisibility on that block. After removing the explicit factor p from the second diagonal block, its coordinate indexed by $0 \le r < N_0$, $0 \le i < e$ has point kernel

$$D_{r,i} = \mathbf{1}_{\{r \equiv 0 \pmod p\}} \mathsf{H}_{\bar{i}}.$$

Indeed the low-digit equation in its defining coefficient packet is

$$ps(x-r) \equiv i - y \pmod{p^2}.$$

If $x - r$ is a nonunit the packet cancels; if it is a unit there are p lifts, whose sum is zero unless $r \equiv 0 \pmod p$, and in the remaining case the packet is exactly $\mathsf{H}_{\bar{i}}$. Since this function is in $\mathcal{C}_2$, every such coordinate pairs to zero with f. This proves $s_{f,2} = 0$ and the lemma. □

Proposition 8.7 (Exact divided-transfer sequence)**.** *For every prime p, let $\pi_* : \mathsf{A}_3 \to \mathsf{A}_2$ sum coefficients on point fibres and let*

$$\mathscr{H}_3^{(2)} = \{\widetilde{y} \in \mathsf{A}_3/p^2\mathsf{A}_3 : B_3\widetilde{y} = 0\}, \qquad \mathcal{J}_2 := \mathcal{C}_2^\perp = \bigcap_{\delta \in \mathbb{P}^1(k)} I_\delta.$$

Put

$$S_3 = (\mathsf{L}_3 : p), \qquad \overline{S}_3 = (S_3 + p\mathsf{A}_3)/p\mathsf{A}_3, \qquad \mathcal{Y}_3 = \overline{S}_3^\perp.$$

Then

$$\widetilde{\mathcal{C}}_3 : \mathscr{H}_3^{(2)} \longrightarrow \mathcal{J}_2, \qquad \widetilde{y} \longmapsto p^{-1}\pi_*\widetilde{y} \pmod p$$

factors through reduction onto $\mathcal{Y}_3$, and the induced map $\mathcal{C}_3 : \mathcal{Y}_3 \to \mathcal{J}_2$ is the transpose of η_2 under the natural finite-field pairings. If $\mathcal{X}_A = \mathcal{X}|_{K_\mathcal{A}}$ and $\mathcal{Q}_p := \operatorname{coker}(\mathcal{X}_A^\vee)$, there is a canonical map

$$\Theta : \mathcal{J}_2 \longrightarrow \mathcal{Q}_p, \qquad f \longmapsto [\Lambda_f], \qquad \Lambda_f(b) = \langle s_{f,1}, b\rangle \quad (b \in K_\mathcal{A}),$$

for which

$$\mathcal{Y}_3 \xrightarrow{\mathcal{C}_3} \mathcal{J}_2 \xrightarrow{\Theta} \mathcal{Q}_p \quad \textit{is exact at } \mathcal{J}_2. \tag{8.19}$$

Furthermore, dualizing the kernel sequence of $\mathcal{X}_A$ gives

$$\mathcal{Q}_p \cong K_{\mathcal{A}\mathcal{X}}^\vee, \qquad [\Lambda_f] \longmapsto \Lambda_f|_{K_{\mathcal{A}\mathcal{X}}}. \tag{8.20}$$

Proof. The source sequence and its finite-field dual are

$$0 \longrightarrow \ker \eta_2 \longrightarrow \mathsf{Q}_2/p\mathsf{Q}_2 \xrightarrow{\eta_2} p\mathsf{Q}_3/p^2\mathsf{Q}_3,$$

$$(p\mathsf{Q}_3/p^2\mathsf{Q}_3)^\vee \xrightarrow{\eta_2^\vee} \mathcal{J}_2 \longrightarrow (\ker\eta_2)^\vee \longrightarrow 0, \tag{8.19a}$$

where $\mathcal{J}_2 \cong (\mathsf{Q}_2/p\mathsf{Q}_2)^\vee$. To realize the first dual space, reduction modulo p maps $\mathscr{H}_3^{(2)}$ onto $\mathcal{Y}_3 \cong (p\mathsf{Q}_3/p^2\mathsf{Q}_3)^\vee$. Indeed, reduction of a mod-p^2 harmonic annihilates both $\mathsf{L}_3/p\mathsf{L}_3$ and every divided relation. Conversely, if $y \in \mathcal{Y}_3$, then $B_3 y = pu$; orthogonality to the divided relations gives $u \in \operatorname{im} \overline{B}_3$, so a choice of z with $\overline{B}_3 z = -u$ makes $y + pz$ harmonic modulo p^2.

By corollary 3.4, for every depth-two point basis vector a one may write $\tau_3(a) = B_3^\mathsf{T} c + px$. Adjunction of fibre summation and transfer shows that $\pi_*\widetilde{y}$ is divisible by p modulo p^2 for $\widetilde{y} \in \mathscr{H}_3^{(2)}$. If two such lifts differ by pz with $B_3 z = 0$ modulo p, then $\langle \pi_* z, a\rangle = \langle z, \tau_3(a)\rangle = 0$ modulo p by the same congruence; hence the raw map depends only on the reduction modulo p. Pairing $\widetilde{\mathcal{C}}_3$ with a depth-two line and using the depth-transition identity puts its image in $\mathcal{J}_2 = \mathcal{C}_2^{\perp}$. Finally,

$$\left\langle \widetilde{\mathcal{C}}_3(\widetilde{y}), a \right\rangle = \frac{\langle \widetilde{y}, \tau_3(a)\rangle}{p} \equiv \langle y, x\rangle \pmod p,$$

which is precisely the dual Bockstein defining $\eta_2^{\vee}$. Consequently $\widetilde{\mathcal{C}}_3$ factors through reduction onto $\mathcal{Y}_3$, and the induced map $\mathcal{C}_3$ is the transpose of η_2 in (8.19a).

For exactness, keep the recursive bases of lemma 8.6. Put $R_{\mathrm{top}} = (N + N_0)e$. The complementary top-shell blocks satisfy

$$Z^{\mathsf{T}} T^{\mathsf{T}} = T^{\mathsf{T}} Z^{\mathsf{T}} = p^3 I_{R_{\mathrm{top}}}$$

(the $n = 3$ specialization of the integral dual identity in [1, Theorem 8.3, equation (26)]). A target $f \in \mathcal{J}_2$ lies in $\operatorname{im} \mathcal{C}_3$ exactly when the prescribed coarse point component pf extends to a mod-p^2 harmonic. Transposing the graph form in lemma 8.6 gives $B_3 = \left(\begin{smallmatrix} B_2 & 0 \\ K^{\mathsf{T}} & Z^{\mathsf{T}} \end{smallmatrix}\right)$. The first block is soluble because $f \in \mathcal{J}_2$; the top-shell block is precisely the requirement that

$$Z^{\mathsf{T}} v \equiv -pK^{\mathsf{T}} f \pmod{p^2}$$

be solvable. Put $r_f = K^{\mathsf{T}} f$; replacing v by $-v$ removes the harmless sign. Multiplying by T^{T}, one direction gives

$$T^{\mathsf{T}} r_f \in pT^{\mathsf{T}} \mathbb{Z}_p^{R_{\mathrm{top}}} + p^2 \mathbb{Z}_p^{R_{\mathrm{top}}}. \tag{8.19b}$$

Conversely, if $T^{\mathsf{T}} r_f = pT^{\mathsf{T}} w + p^2 v$, multiplication by Z^{T} gives $p^3(r_f - pw) = p^2 Z^{\mathsf{T}} v$; cancellation in the torsion-free p-adic lattice yields $pr_f = Z^{\mathsf{T}} v + p^2 w$. Hence (8.19b) is equivalent to the original congruence, in both directions.

The two-chart reduction is

$$\overline{T} = \begin{pmatrix} \mathcal{A} & 0 \\ \mathcal{X} & 0 \end{pmatrix}.$$

By lemma 8.6, the divided adjoint $s_f = (s_{f,1}, 0)$. Therefore $s_f \in \operatorname{im} \overline{T}^{\mathsf{T}}$ exactly when $s_{f,1}$ lies in the combined row space of $\mathcal{A}$ and $\mathcal{X}$. Restriction to $K_{\mathcal{A}}$ kills the first-chart row space, so the remaining condition is exactly $\Lambda_f \in \operatorname{im} \mathcal{X}_A^{\vee}$. For $b \in K_{\mathcal{A}\mathcal{X}}$, expanding s_f in the fixed cyclotomic bases gives the coefficient formula (8.5), hence

$$\Lambda_f(b) = \langle f, \delta_1(b)\rangle.$$

Thus Λ_f is defined on all of $K_{\mathcal{A}}$ before restriction, whereas δ_1 is used only on its typed domain $K_{\mathcal{A}\mathcal{X}}$; no unnamed section is being used. This proves exactness in (8.19).

Finally, dualizing $0 \to K_{\mathcal{A}\mathcal{X}} \to K_{\mathcal{A}} \xrightarrow{\mathcal{X}_A} \operatorname{im} \mathcal{X}_A \to 0$ gives (8.20). The argument uses only the integral incidence lattices, the first transfer congruence, the top-shell dual identity and finite-dimensional duality; no Smith multiplicity or conclusion of theorem 9.1 enters. □

Lemma 8.8 (Divided-trace duality)**.** *For $p \geq 5$, the following are equivalent:*

$$d_p = 0; \qquad \delta_1(K_{\mathcal{A}\mathcal{X}}) = 0; \qquad \mathcal{H}_X(K_{\mathcal{A}\mathcal{X}}) \subset \mathcal{C}_2. \tag{8.21}$$

Proof. Since $\mathcal{C}_3 = \eta_2^\vee$, one has $d_p = 0$ exactly when $\mathcal{C}_3$ is onto. Exactness in proposition 8.7 makes this equivalent to $[\Lambda_f] = 0$ for all $f \in \mathcal{J}_2$. Under (8.20), this says

$$\langle f, \delta_1(b)\rangle = 0 \quad (f \in \mathcal{J}_2, b \in K_{\mathcal{A}\mathcal{X}}).$$

The perfect pairing $\mathcal{J}_2 \times (\mathscr{R}_2/\mathcal{C}_2) \to k$ makes this equivalent to $\delta_1(K_{\mathcal{A}\mathcal{X}}) = 0$. Lemma 8.5 gives $\mathcal{H}_A(K_{\mathcal{A}\mathcal{X}}) \subset \mathcal{C}_2$, so (8.6) gives the final equivalence. □

Theorem 8.9 (Cross-chart divided-carry theorem). *For every prime p, $d_p = 0$; equivalently, (O2) holds.*

Proof. Let $p \geq 5$ and $b \in K_{\mathcal{A}\mathcal{X}}$. By lemma 8.2, $Sb \in K_{\mathcal{A}}$, and lemma 8.5 puts $\mathcal{H}_0(Sb)$ in $\mathcal{C}_{2,\infty}$. Equations (8.8)–(8.9) therefore give $\mathcal{H}_X(b) \in \mathcal{C}_{2,0} \subset \mathcal{C}_2$. Lemma 8.8 yields $d_p = 0$.

The primes 3 and 2 are not inferred from this degree bound. The exact $p = 3$ double-kernel certificate checks all 360 zero-cross Newton labels (the 18 candidate columns have rank 18), hence gives $\varepsilon_1(3,3) = 18$. At $p = 2$ the semisimple sector and division by two are unavailable. A separate full-matrix certificate constructs $\mathcal{A}$ and $\mathcal{A}S$ directly and finds ranks 8 and 9, so (4.2) gives $\varepsilon_1(2,3) = 1$ without using a Smith multiplicity. For both primes, two independent Smith lanes give the complete profile; combining those rows with the modular-rank values a_0 and these independently obtained obstruction values in the defect identity gives $d_2 = d_3 = 0$. The distributed $p = 3$ cross-chart certificate additionally checks the divided-carry membership directly; no separate direct O2-membership claim is made for $p = 2$. The exact inputs, hashes and non-circular deductions are recorded in section C. □

The proof uses the exchange only in the direction (8.7); it does not claim that coordinate swap is an involution on the fixed two-chart splitting. Likewise, (8.8) is an identity after compatible candidate relations are combined, not a columnwise divided identity.

9 Exact assembly and the all-prime Smith profile

Put

$$N_3 = p^6, \qquad \sigma_p = v_p|\mathsf{Q}_3| = \frac{3p^6 - p^5 - p^3 - p}{2}, \tag{9.1}$$

and retain β, γ from (2.5). By lemma 8.1 and theorem 8.9,

$$T_{\geq 3} := m_3 + m_4 + m_5 = a_0 + \varepsilon_1(p,3). \tag{9.2}$$

The symbol $T_{\geq 3}$ is used only for this tail sum; H_p remains the boundary-column count from (6.5).

Rank, total size and weighted order give

$$\begin{aligned}
m_0 &= M_p,\\
m_1 &= 2N_3 - 2M_p + T_{\geq 3} + \beta + 2\gamma - \sigma_p,\\
m_2 &= \sigma_p - N_3 + M_p - 2T_{\geq 3} - \beta - 2\gamma,\\
m_3 &= T_{\geq 3} - \beta - \gamma,\\
m_4 &= \beta, \qquad m_5 = \gamma.
\end{aligned} \tag{9.3}$$

Theorem 9.1 (All-prime depth-three Smith profile). *For $p = 2$ and $p = 3$, respectively,*

$$(m_0, m_1, m_2, m_3, m_4, m_5) = (30, 6, 19, 6, 2, 1), \tag{9.4}$$

$$(m_0, m_1, m_2, m_3, m_4, m_5) = (240, 168, 195, 108, 15, 3). \tag{9.5}$$

For every prime $p > 3$ with $p \equiv 1 \pmod 6$,

$$\boxed{\begin{aligned} m_0 &= \frac{p(p+1)(3p^4+4p^3+3p-1)}{18}, \\ m_1 &= \frac{p(p-1)^2(p+1)(6p^2+2p+1)}{18}, \\ m_2 &= \frac{p(p-1)(2p+1)(6p^3-p^2+9p-2)}{36}, \\ m_3 &= \frac{p(p-1)^2(p+1)(3p^2+4p-1)}{18}, \\ m_4 &= \frac{p(p-1)^2(p+2)}{4}, \\ m_5 &= \frac{p(p-1)}{2}. \end{aligned}} \tag{9.6}$$

For every prime $p > 3$ with $p \equiv 5 \pmod 6$,

$$\boxed{\begin{aligned} m_0 &= \frac{p^2(p+1)^2(3p^2+p+1)}{18}, \\ m_1 &= \frac{p(p+1)(2p-1)(6p^3-7p^2-5p+2)}{36}, \\ m_2 &= \frac{p(12p^5-8p^4+19p^3-12p^2-11p+4)}{36}, \\ m_3 &= \frac{p(p-1)(p+1)(6p^3+2p^2-11p+2)}{36}, \\ m_4 &= \frac{p(p-1)^2(p+2)}{4}, \\ m_5 &= \frac{p(p-1)}{2}. \end{aligned}} \tag{9.7}$$

Equivalently,

$$(\mathsf{Q}_3)_{(p)} \cong \bigoplus_{h=1}^{5} (\mathbb{Z}/p^h\mathbb{Z})^{m_h}. \tag{9.8}$$

Proof. The transfer theorem gives m_4, m_5 and excludes higher exponents. Theorem 3.3 gives m_0, proposition 3.5 gives a_0, theorem 7.3 gives ε_1 for odd primes, and the exact exceptional certificate gives $\varepsilon_1(2,3) = 1$. By theorem 8.9, the defect term in lemma 8.1 is zero. All hypotheses of (9.3) therefore hold for every prime. Direct substitution gives (9.4)–(9.7). No value at $p = 5, 7$ or 11 is used to select either branch. □

The formulas satisfy identically

$$\sum_{h=0}^{5} m_h = p^6, \qquad \sum_{h=1}^{5} h m_h = \frac{3p^6 - p^5 - p^3 - p}{2}. \tag{9.9}$$

After substituting $p = 6h + 1$ in (9.6), every multiplicity expands as a polynomial in $\mathbb{Z}_{\geq 0}[h]$; for example

$$m_2 = 3h(4h + 1)(6h + 1)(216h^3 + 102h^2 + 25h + 2),$$

and the same direct expansion of (9.7) after $p = 6h+5$ lies in $\mathbb{Z}_{\geq 0}[h]$ for all six multiplicities. The public companion performs both substitutions with exact rational arithmetic and rejects a nonintegral or negative coefficient. Thus integrality and positivity do not rely on a bounded prime sweep.

10 Finite controls, source boundary, and limitations

The first profile rows are

p	m_0	m_1	m_2	m_3	m_4	m_5
2	30	6	19	6	2	1
3	240	168	195	108	15	3
5	4050	4140	4795	2490	140	10
7	26740	34608	36225	19488	567	21
11	363000	545622	561979	297330	3575	55

(10.1)

The $p = 2, 3$ rows are reconstructed independently by the public exceptional packages and also agree with the assembled formulas. The $p = 5, 7, 11$ rows are direct exact evaluations of (9.6)–(9.7) and of the tracked structured formula object. All five rows are public regression controls; none is an input used to select a residue-class branch.

The active common-image thresholds at the three hostile odd-prime controls are

p	$\{\tau_p(K) < e\}$
3	$9, 12, 15$
5	$45, 50, 55, 60, 64, 88, 92, 96$
7	$114, 120, 126, 133, 140, 147, 154, 203, 210, 216, 222, 228, 234, 240, 288.$

(10.2)

At $p = 5$, the threshold 64 is a required pattern break: any rule that predicts only an arithmetic progression fails there.

The bounded literature record is not novelty clearance. Molchanov treats the affine residue-ring Radon transform, its normal operator, spectrum and inversion [16]; Ilmavirta gives the finite-group transform context [7]. Adjacent projective Hjelmslev rank and lifting results are due to Landjev–Vandendriessche, Kohnert and Hanssens–Van Maldeghem [6, 8, 12]. Łaba–Trainor and Dvir study the modular hyperplane-function object and rank bounds [5, 10], not the integral affine Smith profile here. Two additional projective-rank sources were available only at metadata or abstract level [2, 11]; no theorem is transferred from them.

The present proof is depth-specific. It does not prove a standard-basis or Rees strictness theorem in general, an all-depth recurrence for closed Smith profiles, or a complexity improvement for the full incidence matrix. The Newton-minors argument succeeds because it computes the actual truncated image directly; it does not justify the false identity $\mathrm{in}(U \cap V) = \mathrm{in}(U) \cap \mathrm{in}(V)$ in arbitrary filtered modules.

11 Public reproducibility and companion interface

The public repository contains the cited-only manuscript source, the structured formula object from which the displayed branches are generated, an exact-arithmetic companion, bounded certificates for the exceptional primes, hostile tests, and fail-closed source and release manifests. The one-command verifier checks:

1. the literal formulas (7.2), (7.7), and (9.4)–(9.7);
2. both polynomial branches of (9.9), integrality substitutions and the finite rows (10.1);
3. the typed distinction among vector G_K, boundary count H_p, generator invariant $\mathrm{ngen}_{\overline{\mathcal{O}}}(W)$ and tail sum $T_{\geq 3}$;
4. the symbolic rank and weighted-order identities for both residue classes modulo 6, together with independent numerical controls;
5. the exact $p = 2$ two-chart rank gap $9-8 = 1$ and its restriction to the 24-dimensional first-chart kernel;
6. source-manifest, PDF, current-checkout and release-surface integrity.

The layout follows the Depth-2 companion [1]: `paper/` contains the source, generated formula fragment and title-named PDF; `companion/` contains the structured formulas and exact arithmetic; `tests/` contains mutation-sensitive controls; `scripts/` supplies the verifier; and `docs/` states source, claim and reproducibility boundaries. Reproduction commands and the hash-locked dependency environment are documented in `REPRODUCE.md`. Finite replay is a supporting check and is not used as a proof of the all-prime theorem.

A Typed symbol contract

The following roles are normative:

$N = p^3,\ N_0 = p^2$	depth-three and coarse residue moduli
$E_{\mathcal{A}}, E_{\mathcal{X}}$	first-chart and cross-chart targets
$\mathtt{q}$	formal Gaussian-polynomial indeterminate
G_K	boundary vector in $\overline{\mathcal{O}}^{p^2}$
H_p	number of admitted boundary columns
$\tau_p(K)$	DVR Smith threshold
$\mathrm{ngen}_{\overline{\mathcal{O}}}(W)$	minimal number of module generators
$T_{\geq 3}$	$m_3 + m_4 + m_5$
$\mathcal{H}_d, \mathcal{H}_X$	first-chart residue and cross divided carries
$\mathcal{C}_{2,\delta}$	depth-two tangent-bundle line code.

In particular, no scalar G_p is defined, and the auxiliary letter H is not used for the Smith tail.

B Residue-block arithmetic for the closed sum

This appendix supplies the complete block decomposition used in lemma 7.2. If $M = ap+b$ with $0 \le b < p$, then

$$\sum_{K=0}^{M-1} \left\lfloor \frac{K}{p} \right\rfloor = p\frac{a(a-1)}{2} + ab. \tag{A.1}$$

For $p = 6h+1$ one has

$$M = 2hp + h = h(2p-1) + 2h, \qquad c = 4h.$$

Thus (A.1), with $(a, b) = (2h, h)$, gives the first floor sum. For the shifted denominator $d = 2p - 1 = 12h + 1$, the interval

$$c \le K + c \le c + M - 1 = hd + 6h - 1$$

has an initial block of length $d - c = 8h + 1$, $h - 1$ complete intermediate blocks, and a terminal block of length $6h$. Hence

$$\sum_{K=0}^{M-1} \left\lfloor \frac{K+c}{d} \right\rfloor = d\sum_{j=1}^{h-1} j + 6h^2 = d\frac{h(h-1)}{2} + 6h^2. \tag{A.2}$$

The ramp in a complete d-block is

$$\sum_{r=p}^{d-1} (r - (p-1)) = \frac{p(p-1)}{2} = 3h(6h+1).$$

It occurs in the initial block and the $h-1$ complete blocks, but not in the terminal block, giving $\sum \delta_K = 3h^2 p$.

For $p = 6h + 5$,

$$M = (2h+1)p + 4h + 3, \qquad d = 12h + 9, \qquad c = 7h + 5.$$

Equation (A.1) with $(a, b) = (2h+1, 4h+3)$ gives the first floor sum. Now

$$c + M - 1 = (h+1)d + 6h + 3.$$

There is an initial block of length $d - c = 5h + 4$, followed by h complete blocks and a terminal block of length $6h + 4 = p - 1$. Consequently

$$\sum_{K=0}^{M-1} \left\lfloor \frac{K+c}{d} \right\rfloor = d\frac{h(h+1)}{2} + (h+1)(p-1). \tag{A.3}$$

The initial ramp omits the values $1, \ldots, h$, each complete block contributes $p(p-1)/2$, and the terminal ramp is zero. Therefore

$$\sum_{K=0}^{M-1} \delta_K = (h+1)\frac{p(p-1)}{2} - \frac{h(h+1)}{2}, \tag{A.4}$$

which is the last entry of (7.9). These decompositions also give $L_{M-1} = p - 1$ and $L_M = 0$ directly.

C Exceptional branches and executable certificates

The primes 2 and 3 are separate proofs, not data used to fit the odd-prime formulas. The executable package is

`companion/exceptional_profiles/`.

It constructs the canonical point-by-line transpose $B_3^\intercal$ and computes its local Smith valuations by two independent exact lanes:

1. minimum-valuation pivot elimination over $\mathbb{Z}/p^7\mathbb{Z}$;
2. Bockstein layer peeling.

Both lanes return

$$\begin{array}{c|c|c|c} p & \text{shape}(B_3^\intercal) & (m_0,\ldots,m_5) & \sum_{h=1}^{5} h\, m_h \\ \hline 2 & 64\times 96 & (30,6,19,6,2,1) & 75 \\ 3 & 729\times 972 & (240,168,195,108,15,3) & 957. \end{array} \tag{B.1}$$

The matrix SHA-256 values are respectively

`CAD9CC122B4811426CE958D1EA9487C536BF51FEACD3550FA04A94A012EB5E82`,
`3D05CC493F837A76693934970005F025F3CC37C70DCECDD659295A37BC23CC68`.

The canonical result and verifier source are authenticated by the public source manifest; normal and optimized executions reproduce the same result bytes.

The same directory contains a separate first-transfer certificate. At $p = 2$ and $p = 3$ it solves exactly for one coarse point-fibre indicator in the modular depth-three line code, checks the solution residual, and then independently pairs the indicator with a basis of the full dual nullspace. The matrix ranks are 30 and 240, the nullities are 34 and 489, and the displayed primal certificates use 14 and 147 lines, respectively. Deleted-line and flipped-point mutations are rejected. Affine translation therefore proves the first-transfer congruence for every point fibre at the two exceptional primes.

The independent package

`companion/epsilon1_p2/`

certifies the obstruction in (4.2) at $p = 2$ without reading (B.1). In the power basis of $\mathbb{F}_2[Z]/(Z^4 + 1)$, its 32 columns are indexed by X^iY^j, $0 \le i < 4$, $0 \le j < 8$, and its rows by (t, b), $0 \le t < 8$, $0 \le b < 4$. It constructs

$$\mathcal{A}_{(t,b),(i,j)} = [Z^b]Z^{i+tj}, \qquad (\mathcal{A}S)_{(t,b),(i,j)} = [Z^b]Z^{ti+j}. \tag{B.2}$$

Two independent exact rank algorithms give

$$\operatorname{rank}_{\mathbb{F}_2} \mathcal{A} = 8, \qquad \operatorname{rank}_{\mathbb{F}_2} \begin{bmatrix} \mathcal{A} \\ \mathcal{A}S \end{bmatrix} = 9, \qquad \dim_{\mathbb{F}_2} K_{\mathcal{A}} = 24. \tag{B.3}$$

Direct restriction gives $\operatorname{rank}((\mathcal{A}S)|_{K_\mathcal{A}}) = 1$; unit-slope rows contribute rank zero on the kernel, while nonunit-slope rows already contribute rank one. The certificate also records an invertible 9×9 witness minor and checks every cyclotomic remainder by both a closed formula and polynomial long division. Thus (4.2) yields $\varepsilon_1(2,3) = 1$ independently of the Smith row.

There is no circular use of theorem 9.1: (B.1) is obtained directly from the incidence matrix. In the defect identity (8.1), the tails from (B.1) are

$$6 + 2 + 1 = 9 = 8 + \varepsilon_1(2,3), \qquad 108 + 15 + 3 = 126 = 108 + \varepsilon_1(3,3),$$

where the independently certified obstruction values are 1 from (B.2)–(B.3) and 18 from the $p = 3$ cross-chart rank. Thus the direct rows imply $d_2 = d_3 = 0$. For $p = 3$ the cross-chart package in `companion/cross_chart_p3/` reconstructs all 378 Newton labels, classifies 360 individually zero cross-chart labels, proves that the remaining 162×18 matrix has rank 18, and checks every divided carry against both the correct line-row space and its annihilator differences. This is an additional local check, not the source of (B.1).

From the repository root, the replay commands are to be run sequentially:

```
python companion/exceptional_profiles/verify_exceptional_profiles.py
python -O companion/exceptional_profiles/verify_exceptional_profiles.py
python companion/exceptional_profiles/verify_o1_exceptional.py
python -O companion/exceptional_profiles/verify_o1_exceptional.py
python companion/cross_chart_p3/verify_cross_chart_p3.py
python -O companion/cross_chart_p3/verify_cross_chart_p3.py
python companion/epsilon1_p2/verify_epsilon1_p2.py
python -O companion/epsilon1_p2/verify_epsilon1_p2.py
```

Each verifier is read-only and authenticates its result and execution receipt. No exceptional computation is extrapolated to another prime.

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